

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	B Tech-AI&DS	Date:	07-08-2026

Code of Conduct

I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Greeshma Reddy

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
 3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.
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CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
 2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
 3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
 4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.
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CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.
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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. MLE of Bigram Probability

Understanding of Case

- The keyboard predicts the next word using a Bigram Language Model.
- MLE estimates probability from corpus frequencies.
- Given:
 - $C(\text{data}) = 3$
 - $C(\text{data science}) = 3$

Application of Concepts

MLE Formula:

Analysis

- Probability **1.0 (100%)** means every occurrence of **data** is followed by **science** in the training corpus.
- Therefore, the keyboard strongly predicts **science** whenever the user types **data**.
- Prediction confidence is very high.

Solution Design

- Store Bigram probabilities in the prediction model.
- Rank **science** as the top suggestion after **data**.
- Update probabilities as more user data is collected.

Clarity

Answer: $P(\text{science} \mid \text{data}) = 1.0$

Interpretation:

The keyboard always predicts **science** after **data**, improving typing speed and accuracy.

2. Backoff Model

Understanding of Case

- The sequence **data science improves** is unseen.
- Since the trigram does not exist, the Backoff Model uses lower-order n-grams.

Application of Concepts

Backoff process:

Trigram:

$P(\text{improves} \mid \text{data science})$

↓

Not found

↓

Bigram:

$P(\text{improves} \mid \text{science})$

↓

Not found

↓

Unigram:

$P(\text{improves})$

Assume the unigram probability is used.

Analysis

- The unseen trigram receives zero probability.
- Backoff avoids zero probability by using lower-order models.
- Thus,

Solution Design

Algorithm:

1. Check trigram.
2. If unavailable → check bigram.
3. If unavailable → use unigram.
4. Predict based on the unigram probability.

Clarity

Backoff ensures unseen word sequences still receive a valid probability, making predictive keyboards more reliable in real-time typing.

3. Deleted Interpolation

Understanding of Case

Deleted Interpolation combines trigram, bigram, and unigram probabilities.

Given

- $\lambda_1 = 0.5$
- $\lambda_2 = 0.3$
- $\lambda_3 = 0.2$

Application of Concepts

Interpolation Formula

Given

Trigram probability

Bigram probability

Unigram probability

Therefore,

Analysis

- Trigram contributes the highest confidence.
- Bigram provides additional support.
- Unigram ensures coverage if higher-order counts are sparse.

Solution Design

Use weighted interpolation for every prediction instead of relying on a single n-gram.

Clarity

Interpolated Probability = 0.932

Interpolation improves prediction accuracy while handling sparse datasets.

4. Entropy Calculation

Understanding of Case

Entropy measures uncertainty in predictions.

Given

- $P(\text{is})=0.66$
- $P(\text{drives})=0.33$

Application of Concepts

Formula

Analysis

- Entropy is less than 1 bit.
- Lower entropy indicates higher prediction confidence.
- The keyboard is more certain about suggesting **is**.

Solution Design

Use entropy to determine whether to show one prediction or multiple suggestions.

Clarity

Entropy ≈ 0.92 bits

The prediction system has relatively low uncertainty and high confidence.

CASE STUDY 2: AI-Powered Customer Support Chatbot

1. Penn Treebank POS Tagging

Understanding of Case

The chatbot assigns POS tags before understanding user intent.

Application of Concepts

Sentence 1

Book/VB

a/DT

flight/NN

ticket/NN

now/RB

Sentence 2

This/DT

book/NN

is/VBZ

interesting/JJ

Analysis

- In Sentence 1, **Book** is an action → Verb.
- In Sentence 2, **book** refers to an object → Noun.

Solution Design

Use contextual words to determine the grammatical role before response generation.

Clarity

Sentence 1

Book/VB a/DT flight/NN ticket/NN now/RB

Sentence 2

This/DT book/NN is/VBZ interesting/JJ

2. HMM Probability

Understanding of Case

The HMM calculates the probability of each possible POS tag.

Application of Concepts

Formula

Similarly

Analysis

Since

$0.30 > 0.20$

Verb is more likely.

Solution Design

Select the tag having the highest probability.

Clarity

Verb Probability = **0.30**

Noun Probability = **0.20**

Therefore **book** is tagged as **VB**.

3. Rule-Based vs HMM Tagging

Understanding of Case

Both methods resolve lexical ambiguity.

Application of Concepts

Rule-Based	HMM
Uses grammar rules	Uses probabilities
Fast	Learns from data
Difficult to maintain	Highly scalable
Less flexible	Handles ambiguity well

Analysis

HMM adapts to different writing styles and unseen text.

Solution Design

Use HMM for large chatbot systems.

Clarity

Recommendation: HMM tagging is better because it is accurate, scalable, and handles ambiguous words effectively.

4. Importance of POS Tagsets

Understanding of Case

POS tags identify grammatical roles.

Application of Concepts

They help the chatbot:

- Detect user intent
- Generate correct responses
- Improve parsing
- Reduce ambiguity

Analysis

Standardized tags provide consistency across different datasets.

Solution Design

Integrate POS tagging into the NLP pipeline before intent classification.

Clarity

POS tagging improves chatbot accuracy, response quality, and language understanding.

CASE STUDY 3: News Analytics and POS Tag Correction System

1. Apply Transformation Rule

Understanding of Case

The Brill Tagger applies learned correction rules after initial tagging.

Application of Concepts

Initial Tags

economic/JJ

growth/NN

increases/NNS

employment/NN

Rule

Change NNS → VBZ if previous tag = NN

Corrected Tags

economic/JJ

growth/NN

increases/VBZ

employment/NN

Analysis

After the noun **growth**, **increases** functions as a singular verb.

Solution Design

Apply transformation rules after statistical tagging.

Clarity

Final POS Tags

economic/JJ growth/NN increases/VBZ employment/NN

2. Error Analysis

Understanding of Case

The initial tagger incorrectly identified **increases** as a plural noun.

Application of Concepts

Sentence structure:

Subject → growth

Verb → increases

Object → employment

Analysis

The corrected tag matches both grammar and sentence meaning.

Solution Design

Use contextual transformation rules to fix tagging mistakes.

Clarity

Transformation-Based Tagging improves grammatical correctness.

3. Frequency Distribution

Understanding of Case

Word frequencies help estimate probabilities.

Total words

Application of Concepts

Word	Frequency	Probability
economic	120	$120/1160 = 0.103$
growth	450	$450/1160 = 0.388$
increases	210	$210/1160 = 0.181$
employment	380	$380/1160 = 0.328$

Analysis

Frequently occurring words receive higher probabilities, improving POS tagging accuracy.

Solution Design

Use frequency counts in probabilistic models such as HMM.

Clarity

Frequency distributions provide prior probabilities that support accurate POS prediction.

4. Transformation-Based Tagging and Entropy

Understanding of Case

Brill Tagging improves the initial statistical output.

Application of Concepts

- Initial tagger makes probability-based predictions.

- Transformation rules correct context-specific errors.
- Entropy measures uncertainty in tag assignments.

Analysis

- Before correction, entropy is higher because multiple tags are possible.
- After correction, entropy decreases as the correct tag becomes more certain.
- Lower entropy indicates higher confidence and improved tagging accuracy.

Solution Design

Use a hybrid approach:

1. Perform initial probabilistic tagging.
2. Apply transformation rules.
3. Evaluate entropy to measure confidence.
4. Produce the final corrected tags.

Clarity

Transformation-Based Tagging improves accuracy by correcting contextual errors, while entropy helps quantify uncertainty before and after rule application. Lower entropy after applying the rule indicates more reliable POS tagging.